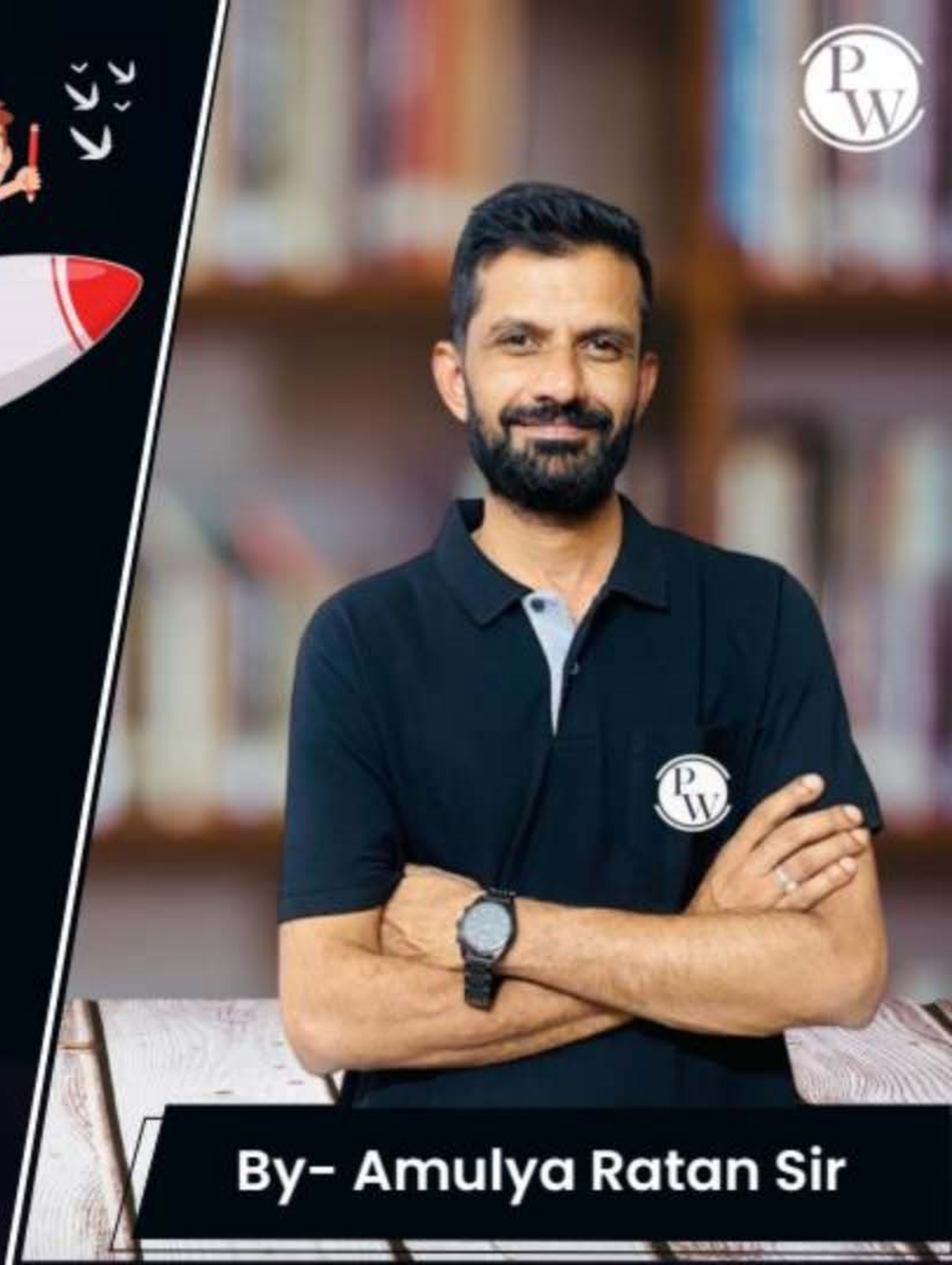
GATE

ALL BRANCHES

General Aptitude

QUANTITATIVE APTITUDE



Lecture No.- 12

By- Amulya Ratan Sir

Recap of Previous Lecture

Topic Counting Theory

Topics to be Covered

Topic-1 Mensuration 2D

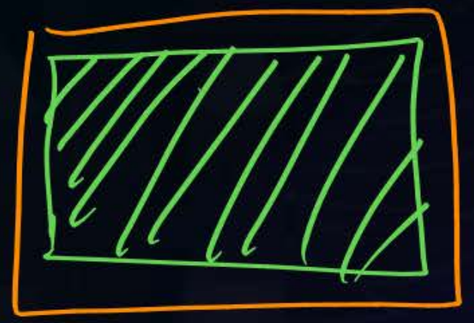
Topic-2 Mensuration 3D

MENSURATION: 2D

Perimeter

Area

1. TRIANGLE
2. QUADILATERAL
3. CIRCLE



Triangles:

Side

1. Scalene △
2. Isosceles △
3. Equilateral △

Angle

(All ∠es) Less than 90° — 1. Acute ∠ed △

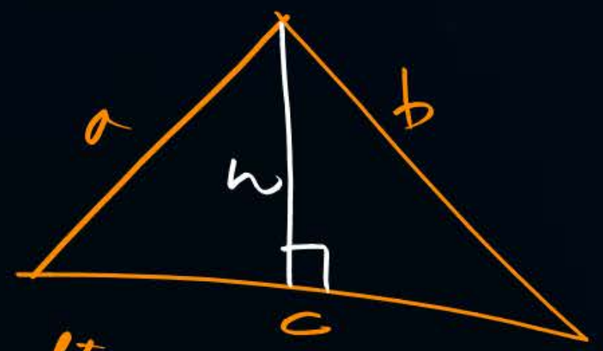
(only one ∠e) More than 90° — 2. Obtuse ∠ed △

one ∠e is 90° — 3. Right ∠ed △

Triangles:

Perimeter (P)

$= a+b+c$

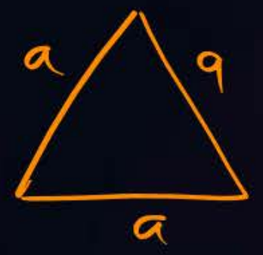


(A) Area $= \frac{1}{2} \times \text{base} \times \text{height}$

Heron's formula

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

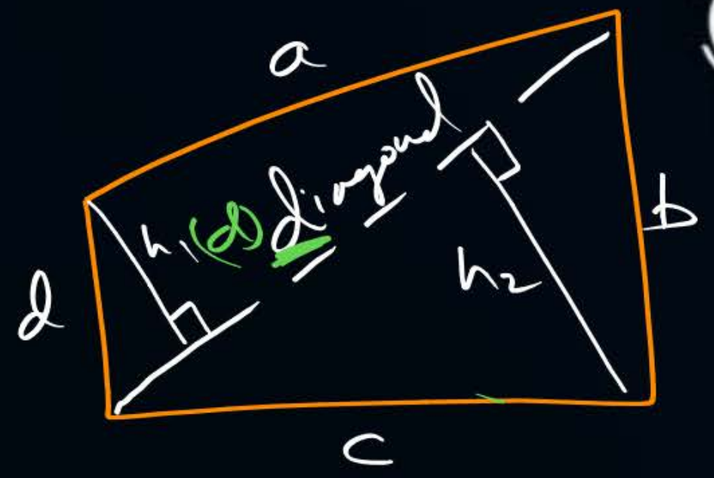
where $s = \frac{a+b+c}{2}$



$P = 3a$

$A = \frac{\sqrt{3}}{4} a^2$

Quadilaterals:



$$P = a + b + c + d$$

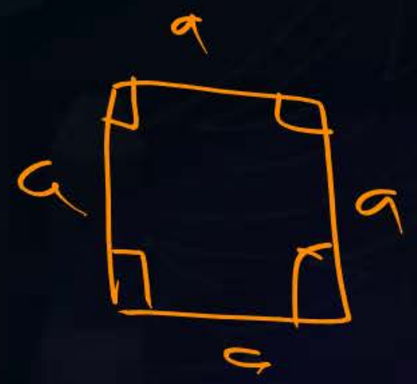
$$A = \frac{1}{2} \times h_1 \times \text{diagonal} + \frac{1}{2} \times h_2 \times \text{diagonal}$$

$$= \frac{1}{2} \times d \times (h_1 + h_2)$$

Rectangle and Square:

$$P = 2l + 2b = 2(l+b)$$

$$A = l \times b$$

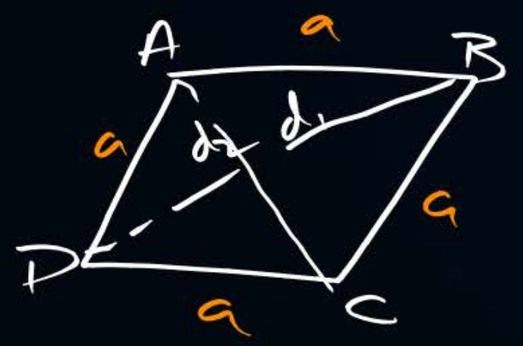


$$P = 4a$$

$$A = a^2 = (\text{side})^2$$

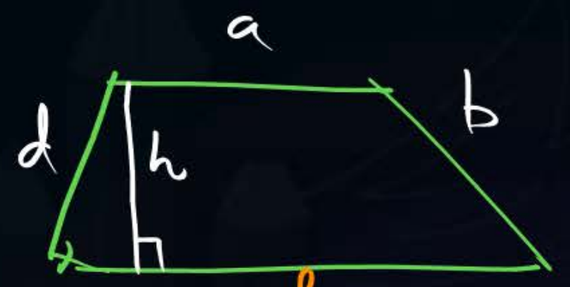
Rhombus, Trapezium & Parallelogram:

$P = 4a$ | $A = \frac{1}{2} \times d_1 \times d_2$



$P = a + b + c + d$

$A = \frac{1}{2} \times h \times (\text{Sum of } \parallel \text{ sides})$



$P = 2(l+b)$

$= \frac{1}{2} \times$

$A = b \times h = \text{base} \times \text{height}$

$AB \parallel DC$

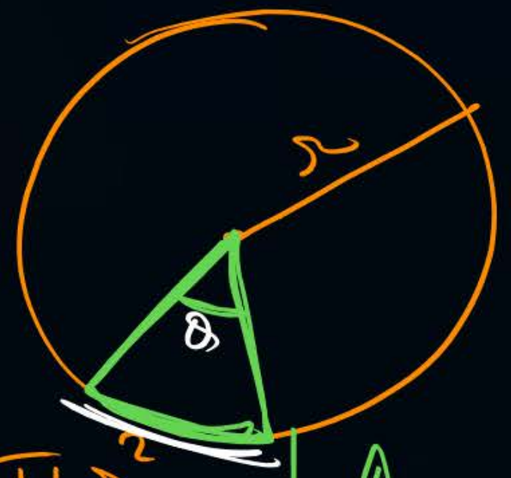
$AD \parallel BC$

Circles & Sectors:

Area (A)

~~Perimeter~~

Circumference (C)



Area of Circle $= \pi r^2$

Circumference of Circle $= 2\pi r$

Area $= \frac{\theta}{360} \times \pi r^2$

Arc of sector

$= \frac{\theta}{360} \times 2\pi r$

MENSURATION: 3D

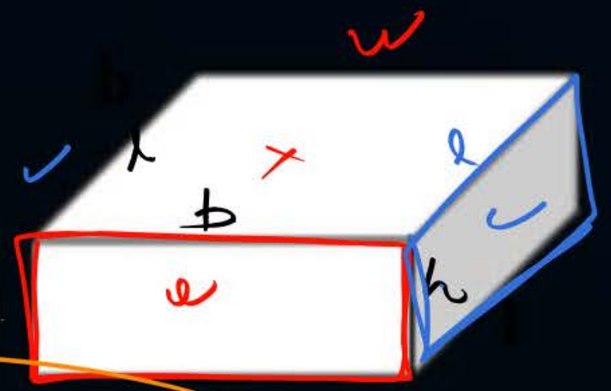
Volume

Lateral Surface Area

Total Surface Area

Cuboid:

(V) Volume $= l \times b \times h$



(LSA) Lateral Surface Area

$= 2bh + 2hl$

$= 2h(l+b)$

Total Surface Area

$= 2bh + 2hl + 2lb$

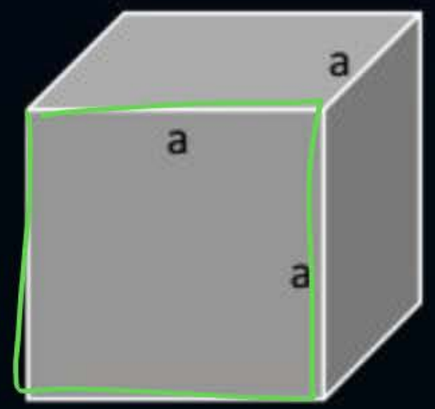
$= 2(lb + bh + hl)$

Cube:

$$\text{Volume} = (a)^3$$

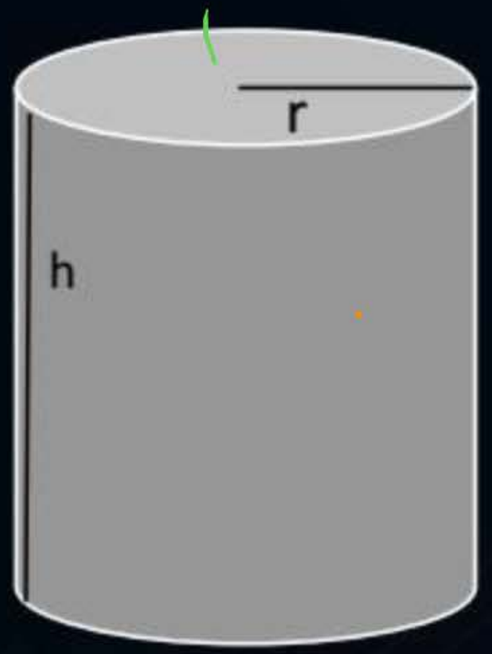
$$LSA = 4a^2$$

$$TSA = 6a^2$$



Cylinder:

Volume $= \pi r^2 h$



Curved Surface Area

$= 2\pi r h$

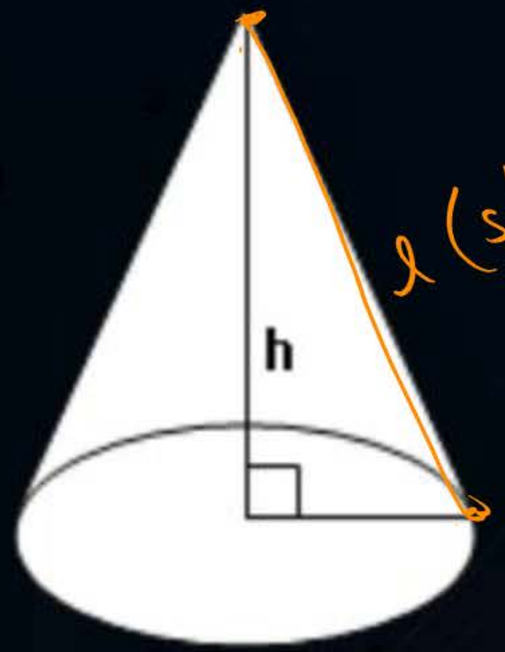
Total Surface Area

$= 2\pi r h + 2\pi r^2$

$= 2\pi r(h + r)$

Cone:

$$\text{Volume} = \frac{1}{3}\pi r^2 h$$



Curved Surface Area

$$= \pi r l$$

Total Surface Area

$$= \pi r l + \pi r^2$$

$$= \pi r (l + r)$$

Sphere:

$$\text{Volume} = \frac{4}{3}\pi r^3$$

$$TSA = 4\pi r^2$$

Hemi-sphere:

$$\text{Volume} = \frac{1}{\cancel{2}} \times \frac{\overset{2}{\cancel{4}}}{3} \pi r^3$$

$$= \frac{2}{3} \pi r^3$$

Curved Surface Area

$$= 2\pi r^2$$

$$TSA = 2\pi r^2 + \pi r^2$$

$$= 3\pi r^2$$

[MCQ]

#Q. What would be the area of sector of a circle whose radius is 12 cm and the length of the arc is 20 cm?

$$\text{Area of Sector} = \frac{\theta}{360} \times \pi r^2$$

A. 60 sq. cm

B. 240 sq. cm

C. 120 sq. cm

D. 64 sq. cm

$$\text{Arc of sector} = \frac{\theta}{360} \times 2\pi r$$

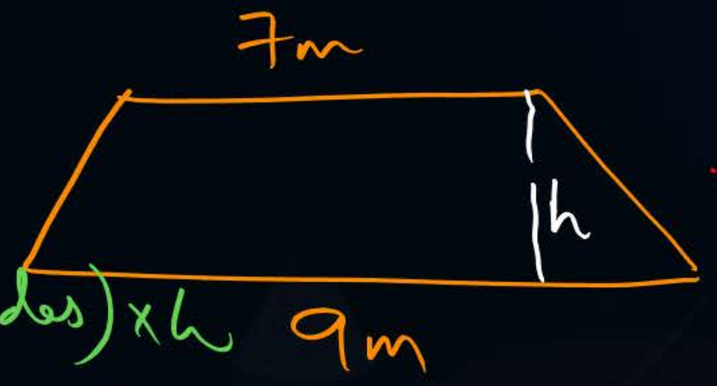
$$\Rightarrow 20 = \frac{\theta}{360} \times 2\pi \times 12$$

$$12 \times 10 = \frac{\theta}{360} \times \pi \times 12 \times 12$$

$$120\,cm^2$$

[MCQ]

#Q. The cross-section of a canal is a trapezium in shape. If the canal is 7 m wide at the top and 9 m at the bottom and the area of cross-section is 128 sq m, find the height of the canal.



Area of Trapezium

$= \frac{1}{2} \times (\text{sum} \parallel \text{sides}) \times h$

A. 32 m

B. 8 m

C. 16 m

D. 4 m

$128 = \frac{1}{2} \times 16 \times h$

$\Rightarrow \frac{128 \times 2}{16} = h = 16\text{ m}$

[MCQ]

#Q. A cube of 6 cm side melted and smaller cubes of 2 cm side are manufactured. Find the number of smaller cubes so formed.

'x'

Volume = a^3

A 12

B 27

C 24

D 8

$6^3 = 2^3 \times x$

$\Rightarrow 216 = 8 \times x$

$\frac{216}{8} = x$

27

[MCQ]

$TSA = 2(lb + bh + hl)$

#Q. The length, breadth & height of a cuboid are in the ratio of 4 : 3 : 2 and its volume is 3000 m^3. Find its total surface area.

20 $l = 4x$ | $b = 3x$ | $h = 2x$ 10

15

A. 1300 m^2

B. 1500 m^2

C. 1333 m^2

D. 2700 m^2

$V = l \times b \times h$

$3000 = 4x \times 3x \times 2x$

$\Rightarrow 3000 = 24 \times x^3$

$\frac{3000}{24} = x^3$

125

$x = \sqrt[3]{125}$ 5

$TSA = 2 \times (300 + 150 + 200)$

$= 2 \times 650$

$= 1300 m^2$

[MCQ]

#Q. The dimensions of a room are 20 m x 15 m x 10 m. What is the cost of painting its four walls at the rate of ₹ 15 per 100 sq m?

l b h

$LSA = 2h(l+b)$

$= 2 \times 10 \times 35$

$= 700 m^2$

A ₹ 70

B ₹ 105

C ₹ 225

D None of these

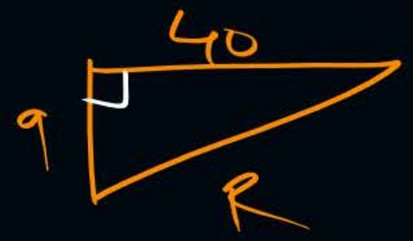
7×15

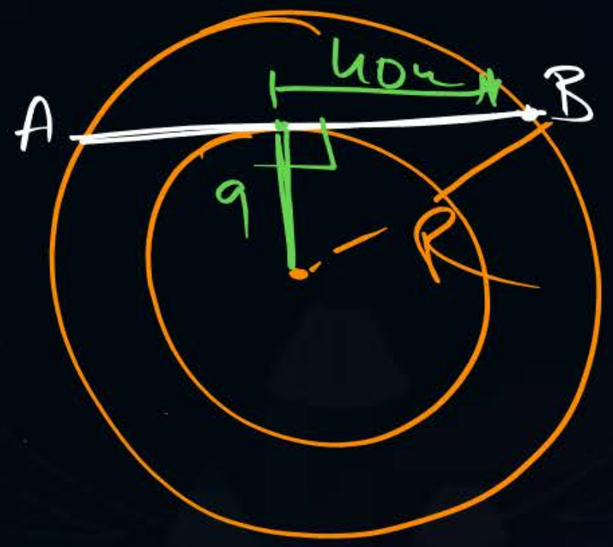
Cost = ₹ 105

[MCQ]

$AB = 80\,cm$

#Q. In two concentric circles, a chord length 80 cm of larger circle becomes a tangent to the smaller circle whose radius is 9 cm. The radius of the larger circle will be?





A 13 cm

B 41 cm

C 52 cm

D 75 cm

$R^2 = 9^2 + 40^2$

$R^2 = 81 + 1600$

$R = \sqrt{1681} = 41\,cm$

[MCQ]

#Q. The diagonal of a rectangle is $\sqrt{41}$ cm and its area is 20 sq cm. The perimeter of the rectangle is?

Assignment

A. 9 cm

B. 18 cm

C. 20 cm

D. 41 cm

2 mins Summary

Topic Mensuration 2D & 3D

THANK - YOU